Introduction to Stata

Lecture 3: Descriptive statistics, tests and graphs

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Where we are, where we go

From Lecture 2 (`data/clean/`)

- `shiw_hh.dta`: households, 2010–2022
- `individuals_full.dta`: persons
- `region_year.dta`: regions

Missing a file? Run `L2_solutions.do`.

Today

- Describe: centre, spread, shape; survey weights
- Tables, simple tests, correlation
- Graphs
- Prepare `shiw_2022.dta` for the regressions of Lecture 4

Handbook: Chapters 6 and 7; § 12.2–12.3.

PART 1

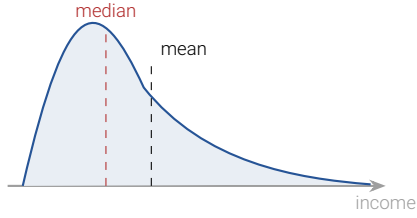
Demonstration

Summary statistics and stored results

Handbook § 6.1, 12.3

```
summarize y_eq if year == 2022, detail
return list
display r(mean) / r(p50)
```

- `detail`: percentiles, skewness, kurtosis
- Results stored in `r()`, overwritten by the next command
- 2022: mean 34,140, median 23,760, skewness 31.7



Income is right-skewed: mean > median. Logs make the distribution closer to symmetric.

Survey weights: sample or population?

Handbook § 6.4

Weight	Meaning	Typical use
`fweight`	each row stands for ***w*** identical rows	aggregated data
`pweight`	inverse probability of being sampled	survey data
`aweight`	row is an average of ***w*** obs. (same means as `pw`)	where `pw` not allowed

```
summarize y_eq if year == 2022                    // sample:     34,140
summarize y_eq [aweight = peso] if year == 2022   // population: 23,027
mean y_eq [pweight = peso] if year == 2022, over(area3)
```

Richer households are over-represented in the 2022 sample: the weights correct for it.

Tables

Handbook § 6.2, 6.3

```
tabulate head_educ area3, column nofreq
tabstat y_eq w_real, by(area3) statistics(n mean p50 sd)
table (area3) (year) [aweight = peso], statistic(mean y_eq)
dtable y_eq_k w_real_k head_age i.head_educ, by(area3, tests) ///
       export("$tables/table1.tex", replace)
```

- `tabulate`: frequencies (`row`, `column`, `chi2`)
- `tabstat`, `table`: statistics by one or two groups
- `dtable` (Stata 18+): a “Table 1” for a paper, exported to `.tex`, `.docx`, `.xlsx`: no copy-paste of numbers, ever

Correlation

Handbook § 6.6

```
correlate ln_y_eq head_educ_years head_age
pwcorr ln_y_eq head_educ_years head_age, ///
       sig star(.05) obs
spearman y_eq w_real
```

- `correlate`: one common N (listwise)
- `pwcorr`: pairwise N, p-values
- `spearman`: ranks, robust to skewness

Pearson correlation measures linear association, has no units, is symmetric.

2022: corr(log income, schooling) = 0.52.

Lecture 4: the OLS slope is a rescaled correlation,

$$\hat{\beta}_1 = \hat{\rho}_{xy} \frac{\hat{\sigma}_y}{\hat{\sigma}_x}.$$

Which test?

Handbook § 6.7

Question	Command	H_0
Do two groups have the same mean?	`ttest y, by(g)`	$\mu_0 = \mu_1$
…without equal variances	`ttest y, by(g) unequal`	$\mu_0 = \mu_1$
Same proportion in two groups?	`prtest d, by(g)`	$p_0 = p_1$
Same mean in $k > 2$ groups?	`oneway y g`	$\mu_1 = \cdots = \mu_k$
Two categorical variables independent?	`tabulate a b, chi2`	independence
Is a correlation zero?	`pwcorr x y, sig`	$\rho = 0$
Same mean, with survey weights?	`mean y [pw=w], over(g)` + `test`	$\mu_0 = \mu_1$

A small p-value says the difference is unlikely to be due to sampling alone. It does not say why the groups differ.

Graphs: one command for each question

Handbook § 7.1–7.6

Question	Command
How is a variable distributed?	`histogram`, `kdensity`
Distribution across groups?	`twoway (kdensity) (kdensity)`, `graph box`
How do means differ across groups?	`graph bar (mean) y, over(g)`
How does a variable change over time?	`twoway line`, `connected`
How are two variables related?	`twoway scatter` + `lfit`, `qfit`
Several graphs together?	`graph combine`

Always: titles and axis labels a reader understands, then `graph export "file.pdf", replace`.

Two graphs from the demonstration

Handbook § 7.1, 7.2, 7.6

```
histogram ln_y_eq if inrange(ln_y_eq, 7, 13), normal ///
    title("Log equivalised income, 2022")
graph export "$figures/hist_log_income.pdf", replace

use "$clean/region_year.dta", clear
twoway (scatter y_eq head_university if year == 2022, mlabel(region_name)) ///
       (lfit y_eq head_university if year == 2022), legend(off)
graph export "$figures/scatter_regions.pdf", replace
```

A few households with income close to zero have very negative logs and stretch the axis: `inrange()` zooms on the central part.

The conditional mean: what a regression tries to capture

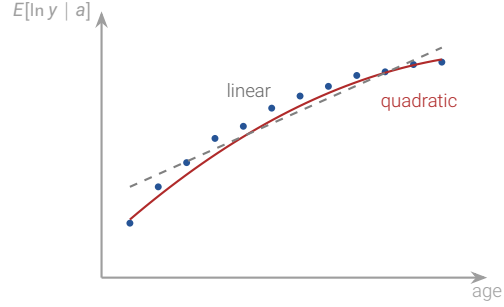
Handbook § 6.3, 7.2

The conditional expectation function

$$E[\ln y \mid \text{age} = a]$$

is the mean of log income among households whose main earner has age a.

- Compute it with `collapse (mean) ln_y_eq, by(age_bin)` (5-year bins: single ages are noisy)
- OLS gives the best linear approximation of this function
- If it is curved, the regression needs it: age^2, logs, groups



Stylised picture; the real one is Exercise 5.

Pre-regression checklist

Check	Why	Commands
Scale	skewed variables: logs (and the zeros)	`summarize, detail`, `ln()`
Outliers	a few extreme values can drive OLS	`_pctile`, winsorizing
Missing values	the regression drops incomplete rows	`misstable summarize`
The sample	same N for descriptives and regressions	`generate sample = ...`
Functional form	is the conditional mean linear?	`collapse`, `twoway`
Collinearity	very correlated regressors	`pwcorr`
Weights	sample or population?	`[pweight = peso]`

Commands of the day

`summarize ..., detail`	distribution	`correlate`, `pwcorr`	correlation
`return list`, `r()`	stored results	`spearman`	rank correlation
`[aweight=]`, `[pweight=]`	survey weights	`ttest`, `prtest`, `oneway`	tests
`mean ..., over()`	weighted means	`histogram`, `kdensity`	densities
`tabulate ..., column chi2`	frequencies	`twoway scatter/lfit`	relationships
`tabstat`, `table`	statistics by group	`graph bar`, `graph box`	groups
`dtable`	“Table 1”	`graph export`	save a graph

PART 2

Now let's practice on Stata

Open `L3_exercises.do`, Part 2: five exercises, 10 minutes each,
then we write the solution together.

Before Lecture 4

Run `L3_solutions.do` (published after this lecture). It rebuilds from the raw data the files that Lecture 4 starts from. If you missed earlier lectures, run all the solutions do-files from Lecture 1, in order.